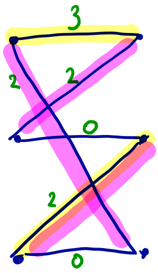
Bipartite Matching and Greedy



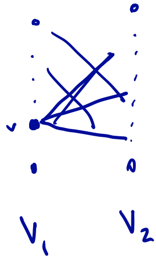
Greedy: 2 edges, weight 5.

Optimal: 3 edges, weight 6

Greedy fails to find a max weight "independent" set.

Bipartite M. as Matroid Intersection

$G = (V_1 \cup V_2, E)$



$M_i = (E, \mathcal{I}_i)$, $i=1,2$

$\mathcal{I}_1 = \{F \subseteq E : \forall v \in V_1 : |F \cap \delta(v)| \leq 1\}$

$\mathcal{I}_2 = \{F \subseteq E : \forall v \in V_2 : |F \cap \delta(v)| \leq 1\}$

partition matroids

Exercise: M_i, $i=1,2$ are indeed matroids

Exercise: $\mathcal{I}_1 \cap \mathcal{I}_2$ = set of all matchings.

$(E, \mathcal{I}_1 \cap \mathcal{I}_2)$ need not be a matroid.

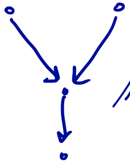
Arborescences

$D = (V, E \subseteq V \times V)$

$T \subseteq E$ is an arborescence if:

(a) No cycles in the undirected subgraph with edges T

(b) Every vertex has at most 1 incoming edge



fails prop (b).

$M_1 = (E, \mathcal{I}_1)$ $\mathcal{I}_1$ is graphic matroid on $G = (V, E)$

$M_2 = (E, \mathcal{I}_2)$ so $I \in \mathcal{I}_1$ is a subset of edges w/o cycles.

$\mathcal{I}_2$ = partition matroid = $\{ F \subseteq E : \forall v \in V, |F \cap \delta^-(v)| \leq 1.$

Exercise: $\mathcal{I}_1 \cap \mathcal{I}_2$ = set of arborescences.

Matroid Intersection

$M_1 = (E, \mathcal{I}_1)$ matroid

$M_2 = (E, \mathcal{I}_2)$ matroid

Matroid intersection: $\mathcal{I}_1 \cap \mathcal{I}_2$.

↳ in general, this is not a matroid.

Study $\mathcal{I}_1 \cap \mathcal{I}_2$:

(a) What is max size of a set in $\mathcal{I}_1 \cap \mathcal{I}_2$?

(b) How do we find this?

(c) Polytope of matroid intersection.

Main Result

Thm : $M_i = (E, \mathcal{I}_i)$, $i = 1, 2$ matroids

with rank functions $r_1 = r_{M_1}$, $r_2 = r_{M_2}$

$$\max_{I \in \mathcal{I}_1 \cap \mathcal{I}_2} |I| = \min_{U \subseteq E} r_{M_1}(U) + r_{M_2}(E \setminus U).$$

Corollaries — 1/2

Corollary: Let M_i as above, and assume $r_1(E) = r_2(E)$.

Then: M_1, M_2 have a common base iff: $r_1(U) + r_2(E \setminus U) \geq r_1(E)$, $\forall U \subseteq E$.

pf: Exercise.

Corollaries — 2/2

Corollary: For M_1, M_2 as above, the minimum size of a common spanning set is:

$$(*)\quad \min_{\substack{S \text{ spans } M_1 \\ \text{and } M_2}} |S| \;=\; \max_{U \subseteq E} \left(r_1(E) - r_1(U) + r_2(E) - r_2(E \setminus U) \right)$$

pf: S must contain a base of M_1 and a base of M_2

$$\Rightarrow |S^*| = \min_{B_1, B_2} |B_1 \cup B_2|$$

$$= \min_{B_1, B_2} |B_1| + |B_2| - |B_1 \cap B_2| = |B_1| + |B_2| - \underbrace{\max_{B_1, B_2} |B_1 \cap B_2|}_{\text{the quantity given by the theorem.}}$$